

NetPad App Summary

What it is

An open-source, enterprise-grade platform for building MongoDB-connected applications that combine forms, workflows, data management, and AI-powered conversational experiences.

Who it's for

Teams building MongoDB-backed internal or customer-facing apps (e.g., ops, product, and engineering) who want rapid form, workflow, and data experiences.

What it does

- Visual form builder with templates, conditional logic, computed fields, and publishing.
- Workflow automation with visual editor, multiple triggers, and queue-based execution.
- Data management: browse collections, query data, import/export, and schema analysis.
- Connection vault for encrypted MongoDB credentials (AES-256-GCM).
- AI experiences: conversational forms, knowledge-guided forms (RAG), and citations.
- Marketplace and npm integration for apps and packages.

How it works (repo evidence)

- Next.js app with API routes under ``src/app/api`` handling org, auth, forms, and workflow endpoints.
- Multi-tenant data model: shared platform DB plus per-organization databases ``org_{orgId}``.
- Connection Vault encrypts connection strings server-side and stores encrypted blobs in MongoDB.
- Form submissions ``insertOne()`` into MongoDB, then explicitly trigger workflow enqueueing.
- Workflow jobs are stored in a MongoDB job queue and processed by ``/api/workflows/process`` via Vercel Cron.
- LLM provider selection uses Ollama when ``OLLAMA_BASE_URL`` is set, otherwise OpenAI.

How to run (minimal)

- Install deps: ``npm install``.
- Create env: ``cp .env.example .env.local`` and set ``MONGODB_URI``, ``SESSION_SECRET``, ``VAULT_ENCRYPTION_KEY``.
- Start dev server: ``npm run dev``.